

L J INSTITUTE OF COMPUTER APPLICATIONS
Master of Computer Applications (MCA)
Syllabus for Batch - 2025
Semester: III

Category	Discipline Specific Core -III			
Course Title	Cybersecurity Fundamentals & Network Security			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	3	0	4	5
Pre-requisites (if any)	- No prior cybersecurity knowledge required – this is the foundation subject - Basic computer literacy and familiarity with internet usage - MCA Sem I & II completed (basic networking concepts from Computer Networks are beneficial) • Lab machines: Wireshark, Nmap, Splunk Free pre-installed; internet access for Shodan lookups			

1. Course Objectives:

CO1	Explain the CIA Triad, types of cyber-attacks, the Cyber Kill Chain, and the distinction between threat, vulnerability, and risk with real-world examples
CO2	Describe the OSI and TCP/IP models at a security-relevant depth: identify which layer each attack and defence mechanism operates at.
CO3	Apply Wire shark and Nmap to capture and analyse network traffic, identify open ports and services, and recognise indicators of suspicious traffic.
CO4	Explain how firewalls, IDS/IPS, VPNs, proxies, and Zero Trust principles defend network infrastructure, and distinguish between them.
CO5	Describe the SOC structure, analyst tier responsibilities, the incident lifecycle, and perform basic log analysis and alert triage using Splunk Free.

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2. Course Contents:

Unit	Course Content	Weightage	Hrs.
Unit I	What is cybersecurity – definition, scope, why it matters in 2025; information security vs cybersecurity distinction; CIA Triad: Confidentiality, Integrity, Availability (with examples); Integrity (hashing, digital signatures); Availability (DDoS, redundancy); core attack types — Malware (virus, worm, ransomware, trojan, spyware), Phishing, Social Engineering, SQL Injection (concept only), DoS/DDoS, Man-in-the-Middle; Threat actors (nation-state, hacktivist, insider, cybercriminal, script kiddie); attack surface concept; Information Security principles: defence in depth, least privilege, separation of duties.	20%	7
Unit II	Non-Tech Student Bridge; OSI Model (all 7 layers with security examples at each); attack mapping to layers — ARP spoofing (L2), IP spoofing (L3), TCP SYN flood (L4), HTTP injection (L7); TCP/IP Model; layers mapped to OSI; characteristics of IPv4 addressing; IPv4 structure, classes, private vs public ranges, subnetting basics (CIDR /24, /16); NAT concept; IPv6 overview; common ports — SSH 22, Telnet 23, SMTP 25, DNS 53, HTTP 80, HTTPS 443, RDP 3389; DNS resolution process step-by-step, record types (A, MX, CNAME, TXT, NS); how HTTPS protects vs what it does not.	20%	7
Unit III	Firewalls — packet filtering, stateful inspection, application-layer (next-gen firewalls); how rules work (allow/deny by IP, port, protocol, direction); DMZ architecture; Intrusion Detection & Prevention Systems — IDS vs IPS, signature-based vs anomaly-based; HIDS vs NIDS; common IDS tools (Snort concept, Suricata overview); how to tune to reduce false positives; VPN site-to-site vs remote access VPN; IPsec and SSL/TLS VPN comparison; split tunnelling; why VPN does not make you anonymous; Proxy Servers — forward proxy vs reverse proxy; Web Filtering vs proxy; transparent vs explicit proxy; how proxies protect internal users; Network Segmentation — flat vs segmented network, VLANs and their security benefit; DMZ design; micro-segmentation concept; why segmentation limits lateral movement; Zero Trust Architecture — never trust always verify principle, identity-centric security; Wireless Security — WPA2 vs WPA3, common Wi-Fi attacks (evil twin, deauth, KRACK concept); securing a wireless network.	20%	8

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Unit IV	Interactive overview; packet capture on live traffic and from pcap file; display filters (ip.addr, tcp.port, http, dns); following a TCP stream; identifying HTTP credentials in cleartext; identifying unusual DNS queries; protocol statistics; Nmap — why port scanning matters; scan types (TCP SYN -sS, TCP connect -sT, UDP -sU); host discovery (-sn); version detection (-sV); OS fingerprinting (-O); NSE script engine intro (--script=default); output formats (-oN, -oX); interpreting results; Network reconnaissance from a defender's view, ethics; on Windows, ss on Linux; common command-line packet tools; reading back, contrast with Wireshark Splunk Free — what is a SIEM, installing Splunk Free, adding a data source; basic SIEM search (index= search by keyword); understanding source types; statistics tab; creating a simple saved search.	20%	7
Unit V	Tier 1 (alert triage, first response); Tier 2 (investigation, deeper analysis); Tier 3 (threat hunting, advanced IR, tuning); SOC Manager; CISO relationship; SOC playbook concept; SIEM in the SOC; log collection from endpoints, firewalls, servers; normalisation and parsing; correlation rules; alerting thresholds; dashboard creation; SIEM use cases (brute force detection, privilege escalation detection, data exfiltration detection); Log Analysis — Windows Event Log structure (Event ID significance — 4624 logon, 4625 failed logon, 4720 account created, 7045 service installed); Linux auth.log, firewall logs; web server access logs; what analysts look for; Alert Handling Process — alert received in SIEM, triage (true positive vs false positive vs escalation criteria, documentation; ticket lifecycle in a SOC tool, Incident Lifecycle: NIST SP 800-61 phases overview — Preparation, Detection & Analysis, Containment, Eradication, Recovery, Post-Incident Review; how Tier 1 hands off to Tier 2; Threat Intelligence basics — IoCs (IP, domain, hash, user agent); threat feeds; how Tier 1 uses TI to contextualise alerts.	20%	7

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Program Outcomes (POs)

PO Code	Program Outcomes
PO1	Computational Knowledge
PO2	Problem Analysis
PO3	Design and Development of Solutions
PO4	Modern Tool Usage
PO5	Individual and Team Work
PO6	Professional Ethics
PO7	Communication Skills
PO8	Life-long Learning

CO-PO Mapping Matrix

Mapping Scale:

- 3 = High
- 2 = Moderate
- 1 = Low

COs \ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	1	2	2	2	2	2	2
CO2.	3	1	2	2	1	1	1	2
CO3	2	2	3	3	2	2	1	2
CO4	3	2	2	2	2	2	1	2
CO5	2	2	3	2	3	2	2	3

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List of Practicals (Samples)

Lab: Wireshark, Nmap, Splunk Free installed. Internet access available for passive OSINT tasks.

Format: each practical = Objective | Environment | Tools | Step-by-step procedure | Expected output | Observation questions | Estimated time

Practical 1 – Wireshark: Capturing and Filtering Live Traffic (Est. time: 3 hrs)

Objective: Capture live network traffic on the lab machine, apply display filters, and identify HTTP credentials and DNS queries.

- Environment: Lab machine with internet access; Wireshark installed
- Tools: Wireshark
- Expected output: Screenshot of TCP Stream showing HTTP request/response; DNS query list screenshot; Protocol Hierarchy statistics screenshot.
- Observation questions: (1) Why can Wireshark see HTTP form data in plaintext? What change to the website would prevent this? (2) Look at the DNS queries – which domains were contacted besides the site you visited? What does this tell you about third-party tracking?

Practical 2 – Nmap: Network Scanning and Service Enumeration (Est. time: 4 hrs)

Objective: Scan the lab's isolated target machine (Metasploitable 2 or a designated scan target) using multiple Nmap techniques and build a complete service inventory.

- Environment: Kali Linux or lab workstation; Metasploitable 2 on isolated host-only network
- Tools: nmap
- Expected output: Completed service inventory table (minimum 8 rows); screenshot of nmap -oN output file; OS detection result.
- Observation questions: (1) You found FTP port 21 open with vsftpd 2.3.4. How would you check if this version has a known vulnerability without exploiting it? (2) What is the difference between a SYN scan (-sS) and a connect scan (-sT)? Which leaves more evidence on the target?

Practical 3 – Analysing a Malicious PCAP File (Est. time: 4 hrs)

Objective: Load a faculty-provided malicious traffic capture and identify the attack type, extract credentials, and build a timeline.

- Environment: Lab machine; faculty-provided malicious.pcap (contains FTP credentials, HTTP SQLi attempt, DNS anomaly)
- Tools: Wireshark
- Expected output: FTP credentials screenshot; SQLi payload found in POST data; DNS anomaly screenshot; completed attack timeline table with minimum 5 events.
- Observation questions: (1) The FTP credentials were visible in plaintext. What protocol would replace FTP to prevent this? (2) What is DNS tunnelling and why is it difficult to detect with a simple firewall rule?

Practical 4 – Firewall Rule Analysis and Network Topology Design (Est. time: 3 hrs)

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Objective: Analyse a given set of firewall rules to identify misconfigurations, and design a secure network topology for a sample organisation.

- Environment: Lab workstation; paper/whiteboard or draw.io
- Tools: draw.io (free, browser-based) or LibreOffice Draw
- Expected output: Corrected firewall rules table (3 rules rewritten with justification); network topology diagram; IDS placement justification paragraph.
- Observation questions: (1) What is the purpose of a DMZ? Why should a web server not sit directly on the internal LAN? (2) A firewall rule says Allow TCP ANY ANY port 443. Why is this dangerous even though 443 is HTTPS?

Practical 5 – Splunk: Log Ingestion and SOC Alert Triage (Est. time: 4 hrs)

Objective: Ingest Windows Security event logs into Splunk Free, write SPL queries to detect brute-force and account creation, and triage a simulated alert.

- Environment: Lab machine with Splunk Free (localhost:8000); faculty-provided Security.evtx and System.evtx log files
- Tools: Splunk Free web UI
- Expected output: Screenshots of all 5 SPL queries with results; 1-paragraph triage note; total failed login count recorded.
- Observation questions: (1) Event ID 4625 with Logon Type 3 vs Logon Type 10 – what is the difference in attack method? (2) If you confirmed this is a brute-force true positive, what are the immediate containment steps a Tier 1 analyst should take?

Practical 6 – Identifying Suspicious Network Traffic Patterns (Est. time: 3 hrs)

Objective: Apply combined Wireshark and Nmap skills to identify suspicious indicators in a lab scenario that simulates a compromised machine phoning home.

- Environment: Kali Linux; a second VM configured by faculty to generate periodic outbound connections to a non-standard port
- Tools: Wireshark, Nmap, netstat
- Expected output: Wireshark screenshot of beaconing pattern; netstat output with suspicious connection highlighted; VirusTotal lookup result; incident note.
- Observation questions: (1) What is Command and Control (C2) beaconing and why do attackers use regular intervals? (2) How could an attacker use port 443 (HTTPS) to hide C2 traffic from a firewall?

Capstone – SOC Monitoring Mini-Project (Weeks 11-13, 8 hrs)

Objective: Build a functional Splunk dashboard that a Tier 1 SOC analyst could use to monitor a Windows environment, using faculty-provided log files simulating one week of activity from a company that experienced a security incident.

Deliverable: A Splunk dashboard with minimum 4 panels: (1) Failed login attempts over time (timechart), (2) Top 10 source IPs in failed logins (bar chart), (3) New accounts created this week (table), (4) Services installed this week (table). Plus a 1-page Incident Report: What happened? When?

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Which account was compromised? What evidence supports your conclusion? What remediation do you recommend?

Assessment: 30 marks – Dashboard accuracy and SPL query quality (15), Incident report (10), Presentation to faculty (5).

Reference Books:

1. "CompTIA Security+ Study Guide" – Mike Chapple & David Seidl, Sybex
2. "Security Operations Center: Building, Operating and Maintaining Your SOC" – Joseph Muniz, Gary McIntyre, Nadhem AlFardan
3. "Network Security Essentials" – William Stallings, Pearson
4. "The Practice of Network Security Monitoring" – Richard Bejtlich, No Starch Press
5. Splunk Documentation – <https://docs.splunk.com>
6. Wireshark User Guide – https://www.wireshark.org/docs/wsug_html_chunked/

Web Resources:

1. TryHackMe Pre-Security Learning Path – <https://tryhackme.com/path/outline/presecurity>
2. Cisco Networking Academy – Introduction to Cybersecurity (free) – <https://skillsforall.com>
3. Google Cybersecurity Certificate – Coursera
4. Cybrary SOC Analyst course – <https://www.cybrary.it>